

VCU Discrete Mathematics Seminar

Backgammon Parking Functions

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Wednesday, September 23
1:00-1:50 EST

On Zoom with 4145 Harris watch party option:

<https://vcu.zoom.us/j/81475528886>
password=graphs2357



We introduce a new parking scheme inspired by the game of backgammon. We consider a one-way street with n parking spots and a queue of n cars waiting to park.

Each car has a preferred parking spot on the street and enters in sequence to park in that spot if it is available. If a car finds its preferred spot occupied, it bumps the already parked car out of that spot and takes its place. The bumped car then returns to the start of the street and parks in the first available spot it encounters. Consequently, all the cars park.

For a given preference list $\alpha \in [n]^n$, the final order in which the cars park is some permutation in $\mathfrak{S}_n$, which we call the outcome of α . One key question is to determine how many preference lists yield a particular outcome. We show that this is bounded above by the Catalan number C_n , and determine all outcomes, including the identity permutation, which attain the maximum. Additionally, we show that the number of preference lists yielding the decreasing permutation $n \cdots 321$ is the Motzkin number M_n .

Joint work with Pamela E. Harris and Maryam Khaqan.

For the DM seminar schedule, see:

<https://go.vcu.edu/discrete>